Hindi Ambiguity Parser

Transformer-based Ambiguity Detection in Hindi Language

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NLP Course Assignment Report

Model: google/mt5-small | ~300M Parameters | Google Colab T4 GPU

1. Introduction

Natural Language Processing (NLP) is the field of artificial intelligence that enables machines to understand, interpret, and generate human language. One of the most fundamental and challenging problems in NLP is **linguistic ambiguity** — the phenomenon where a word, phrase, or sentence can have more than one interpretation.

This project addresses the question: ***How can a Transformer-based language model be trained to detect and resolve ambiguity in the Hindi language?***

We fine-tune **google/mt5-small** — a multilingual Transformer with approximately 300 million parameters — on a hand-crafted dataset of 25 ambiguous Hindi sentences. Given any ambiguous input, the model outputs **two different interpretations**, making the ambiguity explicit and understandable.

1.1 Problem Statement

Assignment Question: Discuss ambiguity in NLP. Train a Transformer (LLM) to handle the ambiguity in the Hindi language.

For example, consider the Hindi sentence:

> "सोना अच्छा है।"

This sentence is ambiguous because the word **सोना** has two meanings:

- सोना = Gold (स्वर्ण) → Gold is precious
- सोना = Sleeping (नींद लेना) → Sleeping is good for health

Our model takes this sentence as input and generates both interpretations automatically.

2. Types of Ambiguity in NLP

Ambiguity in natural language can be classified into several distinct types. This project focuses on four major categories, with particular emphasis on Hindi.

2.1 Lexical Ambiguity (शाब्दिक अस्पष्टता)

Lexical ambiguity occurs when a single word has multiple dictionary meanings. It is the most common type of ambiguity.

Hindi Sentence	Ambiguous Word	Meaning 1	Meaning 2
सोना अच्छा है।	सोना	Gold (स्वर्ण)	Sleeping (नींद)
वह बैंक गया।	बैंक	Financial bank	River bank
आम आदमी खुश है।	आम	Common/ordinary	Mango
दिल टूट गया।	टूट गया	Emotionally broken	Medically damaged

2.2 Syntactic Ambiguity (वाक्यात्मक अस्पष्टता)

Syntactic ambiguity arises when the grammatical structure of a sentence allows for multiple parse trees, leading to different interpretations.

Example: "मैंने उड़ते पक्षी को देखा।"

- Parse 1: I saw [a bird that was flying]
- Parse 2: While [I was flying], I saw a bird

2.3 Referential Ambiguity (संदर्भात्मक अस्पष्टता)

Referential ambiguity occurs when a pronoun or referring expression could point to multiple entities mentioned earlier in the text.

Example: "राम और श्याम मिले, वह खुश था।"

- The pronoun "वह" could refer to Ram or Shyam

2.4 Pragmatic Ambiguity (व्यावहारिक अस्पष्टता)

Pragmatic ambiguity occurs when the literal meaning and the intended communicative purpose of a sentence differ based on social context.

Example: "यहाँ बहुत ठंड है।"

- Literal interpretation: The temperature here is very low (stating a fact)
- Pragmatic interpretation: Please close the window / turn on the heater (implied request)

3. Model Architecture — mT5-small

We use **google/mt5-small**, a multilingual variant of the T5 (Text-To-Text Transfer Transformer) architecture developed by Google Research.

3.1 Why mT5?

- Pre-trained on 101 languages including Hindi
- Encoder-Decoder architecture ideal for Seq2Seq tasks
- ~300M parameters — fits in Google Colab free tier
- Multilingual C4 (mC4) corpus training ensures Hindi linguistic knowledge
- Well under the 2 Billion parameter limit specified

3.2 Architecture Details

Property	Value
Architecture	Encoder-Decoder Transformer
Total Parameters	556,291,456 (~556.3M)
Encoder Layers	6 layers with Self-Attention
Decoder Layers	6 layers with Masked Self-Attention + Cross-Attention
Pre-training Data	mC4 — Multilingual C4 corpus (101 languages)
Task Format	Seq2Seq — Text-in, Text-out
Device	CUDA (Tesla T4 GPU)

3.3 The Attention Mechanism

The core innovation of the Transformer is the **Self-Attention mechanism**. For every word in a sentence, attention computes a relevance score against every other word. This allows the model to understand long-range dependencies — for example, knowing which noun a pronoun refers to even when they are far apart.

In mathematical terms, attention is computed as:

> Attention(Q, K, V) = softmax(QK^T / sqrt(d_k)) * V

where Q (Query), K (Key), and V (Value) are learned linear projections of the input embeddings.

4. Dataset

4.1 Dataset Construction

We manually constructed a dataset of **25 ambiguous Hindi sentences**, each carefully chosen to represent a specific type of linguistic ambiguity. The dataset was designed with the following criteria:

- Each sentence must be genuinely ambiguous in Hindi
- Both interpretations must be plausible and distinct
- All four ambiguity types must be represented
- Sentences cover everyday conversational Hindi

4.2 Dataset Structure

Each entry in the dataset contains the following fields:

Field	Description
hindi_sentence	The ambiguous Hindi input sentence
ambiguity_type	Category: Lexical, Syntactic, Referential, or Pragmatic
ambiguous_word	The specific word or phrase causing ambiguity
interpretation_1	First possible meaning of the sentence
interpretation_2	Second possible meaning of the sentence

4.3 Distribution

Ambiguity Type	Count	Example
शाब्दिक (Lexical)	14	सोना = gold OR sleeping
वाक्यात्मक (Syntactic)	4	उड़ते — who was flying?
व्यावहारिक (Pragmatic)	4	यहाँ ठंड है — fact OR request
अर्थपरक (Semantic)	2	वह चला गया — left OR died
संदर्भात्मक (Referential)	1	वह — Ram or Shyam?
Total	25	

4.4 Train-Test Split

The dataset is split 80/20:

- Training set: 20 sentences
- Test set: 5 sentences

This split is performed using a fixed random seed (42) for reproducibility.

5. Methodology

5.1 Task Formulation

The problem is formulated as a **Sequence-to-Sequence (Seq2Seq)** task. Every NLP task can be cast as text-in, text-out — this is the core philosophy of T5.

Input format:

> "अस्पष्टता विश्लेषण: <hindi_sentence>"

Output format:

> "अर्थ 1: <interpretation_1> [SEP] अर्थ 2: <interpretation_2>"

The **[SEP]** token is a custom separator used to split the two interpretations. After generation, the output is split on [SEP] to extract the two individual interpretations.

5.2 Tokenization

mT5 uses **SentencePiece** tokenization — a sub-word tokenization method that handles rare words and morphologically rich languages like Hindi effectively.

- Max input length: 128 tokens
- Max output length: 256 tokens
- Padding tokens in labels are replaced with -100 so the loss function ignores them

5.3 Training

We use the **Seq2SeqTrainer** from Hugging Face with the following hyperparameters:

Hyperparameter	Value	Purpose
num_train_epochs	10	Full passes through training data
learning_rate	5e-4	Step size for gradient descent
per_device_train_batch_size	4	Sentences processed per update
warmup_steps	10	Gradual LR increase at start
weight_decay	0.01	L2 regularization (anti-overfitting)
fp16	True	16-bit precision — 2x faster on GPU

5.4 Inference

During inference, the model uses **Beam Search** with 4 beams. Beam Search maintains the top-4 partial sequences at each generation step and selects the best complete sequence — producing higher quality output than simple greedy decoding.

A **garbage detection** mechanism is also implemented: after decoding, the output is checked for Hindi Unicode characters (range \u0900–\u097F). If fewer than 5 Hindi characters are found, the model output is considered garbage and the correct interpretation is retrieved directly from the training dataset as a fallback.

6. Implementation

6.1 Environment

- Platform: Google Colab (free tier)
- GPU: Tesla T4 (CUDA enabled)
- Python: 3.10+
- Key Libraries: transformers 5.0.0, datasets 4.0.0, torch 2.10.0+cu128

6.2 Key Code Components

Data Formatting

```
input_text  = f'अस्पष्टता विश्लेषण: {sentence}'
target_text = f'अर्थ 1: {interp1} [SEP] अर्थ 2: {interp2}'
```

Model Loading

```
tokenizer = AutoTokenizer.from_pretrained("google/mt5-small")
model = MT5ForConditionalGeneration.from_pretrained("google/mt5-small")
```

Output Parsing

```
parts   = decoded.split("[SEP]")
interp1 = parts[0].replace("अर्थ 1:", "").strip()
interp2 = parts[1].replace("अर्थ 2:", "").strip()
```

Garbage Fix

```
decoded = re.sub(r"<0x[0-9A-Fa-f]+>", "", decoded).strip()
hindi_chars = re.findall(r"[\u0900-\u097F]", decoded)
if len(hindi_chars) < 5: # fallback to RAW_DATA
```

6.3 Known Issues

mT5-small sometimes outputs raw **SentencePiece byte tokens** (e.g., <0x03>) instead of Hindi text. This is a known tokenizer artifact. Our fix strips these tokens using a regular expression and falls back to ground truth data when necessary.

7. Results

7.1 Training Results

Metric	Value
Final Training Loss	0.0000
Training Time	3244.9 seconds (~54 minutes)
GPU	Tesla T4 (CUDA)
Total Parameters	556,291,456 (556.3M)
Trainable Parameters	556,291,456 (all)

7.2 Sample Output

Input: "सोना अच्छा है।"

- Interpretation 1: सोना (स्वर्ण) बहुत कीमती है। — Gold is very precious.
- Interpretation 2: सोना (नींद लेना) स्वास्थ्य के लिए अच्छा है। — Sleeping is good for health.

Input: "वह बैंक गया।"

- Interpretation 1: वह नदी के किनारे गया। — He went to the riverbank.
- Interpretation 2: वह पैसे जमा करने बैंक गया। — He went to the financial bank.

7.3 Analysis

A final training loss of 0.0000 indicates that the model has **perfectly memorised** the 20 training sentences. With only 25 total samples, this is expected and acceptable for a proof-of-concept. The model performs well on all training-set sentences and demonstrates the feasibility of using a fine-tuned Transformer to generate dual interpretations for ambiguous Hindi input.

8. Conclusion

This project successfully demonstrates the use of a fine-tuned Transformer model to detect and resolve linguistic ambiguity in Hindi. The key contributions are:

1. A hand-crafted dataset of 25 ambiguous Hindi sentences spanning 4 ambiguity types
2. A Seq2Seq task formulation where the model outputs two interpretations separated by [SEP]
3. Fine-tuning of google/mt5-small (~300M params) achieving zero training loss
4. A garbage-detection and fallback mechanism to ensure reliable output
5. An interactive Colab demo for live ambiguity analysis

8.1 Limitations

- Small dataset (25 sentences) — model may not generalize well to unseen sentences
- mT5-small sometimes generates garbled output (mitigated by fallback mechanism)
- No evaluation metric (BLEU, ROUGE) due to dataset size

8.2 Future Work

- Expand dataset to 500+ Hindi sentences for better generalization
- Experiment with larger models (mT5-base or mT5-large)
- Add automatic ambiguity type classification
- Evaluate using BLEU score and human evaluation
- Deploy as a web application or API

9. References

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